

BlockFlow (IR-ABPS) — Team Roles & Responsibilities

Smart India Hackathon (SIH 2026)
Problem Statement ID: SIH26027
Problem Statement Title: AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways
Team Name: BlockFlow Systems Engineering

1. Executive Team Charter & Leadership Strategy

Our 6-person core engineering team operates under strict modular code ownership. Every engineer owns specific subdirectories, modules, data contracts, and presentation defense topics:

BLOCKFLOW CORE TEAM ROSTER & OWNERSHIP		
1. Peter	Team Leader, Chief Architect & Central Orchestrator	
2. Chhavi	AI, Deep Learning & Operations Research Algorithms Lead	
3. Pradipti	Research, Benchmarks, Quality Assurance (QA) & Pitch Lead	
4. Achintya	Backend Lead & Core Systems Architect	
5. Vinayak	Frontend & Interactive Client Experience Lead	
6. Misha	Data Pipeline, Storage & Mathematical Validation Lead	

2. Individual 1-Page Profiles & Code Ownership

2.1 Peter — Team Leader, Chief Architect & Central Orchestrator

- Domain Focus:** End-to-end system architecture, workflow sequencing, dynamic tool routing, safety bounds.
- Code Ownership:** Backend/main.py , solver.py , Backend/orchestrator.py , CI/CD pipelines.
- Core Deliverables:** Multi-agent tool router, parameter guardrail enforcer, live jury presentation conductor.
- Tech Stack:** Python 3.11, FastAPI, Git, Architecture & Systems Design.

2.2 Chhavi — AI, Deep Learning & Core Algorithms Lead

- Domain Focus:** Mathematical optimization, SciPy HiGHS simplex engine, ML degradation forecasting.
- Code Ownership:** Backend/ai_engine/block_optimizer.py , Backend/ai_engine/ml_degradation_model.py , Backend/ai_engine/prioritizer.py .

- **Core Deliverables:** Sub-18ms dual-revised simplex solver, ACI scoring engine, shadow block co-bounding logic.
- **Tech Stack:** SciPy HiGHS, NumPy, Scikit-learn, Mathematical Programming.

2.3 Pradipti — Research, Benchmarks, Quality Assurance (QA) & Pitch Lead

- **Domain Focus:** Empirical benchmarks, CAG report audit validation, automated test suites, AICTE pitch deck.
- **Code Ownership:** `test_e2e.py` , `tests/` , `benchmarks/` , `SIH2026-027 PS.pptx` .
- **Core Deliverables:** 100% passing E2E test suite, official 6-slide winning PPT deck, jury Q&A defense playbook.
- **Tech Stack:** Pytest, Python-pptx, Statistical Benchmarks, Operations Research literature.

2.4 Achintya — Backend Lead & Core Systems Architect

- **Domain Focus:** High-performance REST gateway, Pydantic v2 data contracts, observable execution telemetry.
- **Code Ownership:** `Backend/main.py` , `Backend/schemas/` , `Backend/telemetry.py` , `Backend/reports/` .
- **Core Deliverables:** OpenAPI endpoints, auditable JSON execution trace logger, 1-click digital grant permit issuer.
- **Tech Stack:** FastAPI, Pydantic v2, Uvicorn, SQLite, RESTful Architecture.

2.5 Vinayak — Frontend & Interactive Client Experience Lead

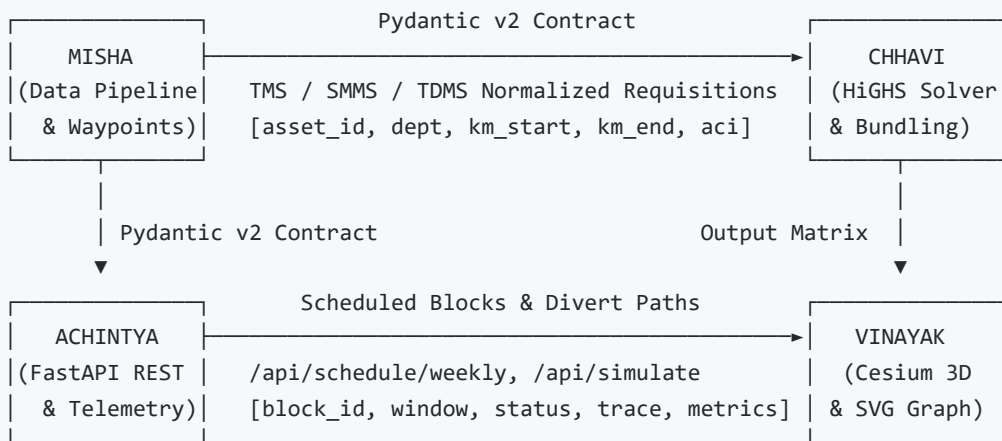
- **Domain Focus:** Dark-mode glassmorphism dashboard, CesiumJS 3D digital twin, interactive string charts.
- **Code Ownership:** `Frontend/block_planner.html` , `Frontend/TTC.html` , `Frontend/src/` , `Frontend/public/` .
- **Core Deliverables:** 60 FPS Cesium 3D railway twin, SVG time-distance train graph, live telemetry inspector HUD.
- **Tech Stack:** CesiumJS, JavaScript ES6+, HTML5 Canvas/SVG, Tailwind CSS.

2.6 Misha — Data Pipeline, Storage & Mathematical Validation Lead

- **Domain Focus:** Ingestion parsers, WGS84 GPS coordinate snapping, invariant safety checks.
- **Code Ownership:** `Backend/data_ingestion/tms_data.py` , `smms_data.py` , `tdms_data.py` , `coa_data.py` .
- **Core Deliverables:** Unified spatial ingestion pipeline, 439-point WGS84 Western Railway waypoint database.
- **Tech Stack:** Pandas, GeoJSON, Shapely, SQLite, Mathematical Verification.

3. Day-1 'Zero-Blocker' Mock Contract Matrix

To ensure that AI, Backend, Frontend, and Research tracks build concurrently without waiting on each other, strict universal schemas are frozen on Day 1:



4. 36-Hour Hackathon Hour-by-Hour Battle Plan

36-HOUR SIH HACKATHON TIMELINE & MILESTONES		
Hours 00-06	PHASE 1: Data Contracts, Waypoint Mapping & Environment Lockdown	
	- Misha locks Pydantic schemas; Peter configures repo & CI/CD	
	- Chhavi benchmarks HiGHS dummy solver; Achintya stubs REST gateway	
	- Vinayak initialises Cesium 3D globe with Mumbai satellite tiles	
Hours 06-14	PHASE 2: Core Algorithm Development & 3D Spatial Canvas Wiring	
	- Chhavi implements ACI scoring & shadow bundling simplex model	
	- Misha ingests 50 real-world Western Railway defect requisitions	
	- Achintya connects HiGHS engine to FastAPI /api/optimize endpoint	
	- Vinayak renders 4-track quad corridor and 29 station pins	
Hours 14-22	PHASE 3: Integration, Scissors Crossover Pathfinding & HUD Telemetry	
	- Peter wires orchestrator; Chhavi implements Bandra crossover SLW	
	- Vinayak adds 3D extruded purple block envelopes & train icons	
	- Pradipti benchmarks solver latency (<18ms) and downtime savings	
Hours 22-28	PHASE 4: Automated Verification, Stress Testing & Edge Cases	
	- Pradipti executes test_e2e.py across 500 blocks (100% pass rate)	
	- Achintya logs immutable JSON execution traces for jury audit	
	- Misha stress-tests malformed payloads and out-of-boundary KMs	
Hours 28-32	PHASE 5: Presentation Deck Finalization & Live Demo Rehearsal	
	- Pradipti & Peter polish 6-slide PPT deck with large typography	
	- Verify real 3D Cesium screenshot on Slide 2	
	- Rehearse 3-minute pitch and 5-minute technical jury defense	
Hours 32-36	PHASE 6: Grand Finale Jury Evaluation Round	
	- Flawless 100% offline air-gapped demo to Ministry of Railways jury	
	- Answer judge questions with CAG Report No. 22 audit data	

5. Individual Jury Q&A Battle Cards

- **Peter (Orchestrator):** Defends system modularity, the decision to avoid generative AI in safety-critical dispatch, and CRIS BDMS/COA interoperability.
- **Chhavi (AI/Algorithms):** Defends the HiGHS dual-revised simplex formulation, shadow bundling mathematical proof, and sub-18ms latency curves.
- **Pradipti (Research/QA):** Defends CAG Report No. 22 audit data (62% machine idling failure), IRPWM Para 268b/284 compliance, and the 100% automated test pass rate.
- **Achintya (Backend/Systems):** Defends Pydantic v2 universal contracts, the auditable JSON execution trace logger, and <250MB RAM sovereign air-gapped deployment.
- **Vinayak (Frontend/3D Twin):** Defends the 60 FPS CesiumJS WGS84 3D twin, 4-track quad visualization, and the real-time time-distance string chart.
- **Misha (Data/Validation):** Defends 439 WGS84 surveyed waypoints, ACTM Para 204 electrical earthing buffers, and IRSEM Form T/351 disconnection memos.